

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations
 December 2023

IS1101-1 – FUNDAMENTALS OF CYBER SECURITY

(For ACT, AI&DS, AI&ML, CB, CCE, CSE, ECE, EEE, ISE, R&AI, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What does the Indian Information Technology Act (ITA-2008) aim to protect?**
 A) Only data
 B) Only systems
 C) Both data and systems
 D) Neither data nor systems
2. **What is the main goal of Password Sniffing?**
 A) Altering data
 B) Financial gain
 C) Stealing passwords
 D) Vandalizing websites
3. **Which of the following is a form of Techno-vandalism?**
 A) Email Spoofing
 B) Website Defacement
 C) Data Diddling
 D) Salami Attack
4. **What does Software Piracy involve?**
 A) Unauthorized copying
 B) Stealing passwords
 C) Defacing websites
 D) Sending fake emails
5. _____ gets propagated through networks and technologies like SMS, Bluetooth, wireless medium, USBs and infrared to affect mobile phones.
 A) Worms
 B) Antivirus
 C) Malware
 D) Multimedia files
6. **Which of the following is not a security issue for PDAs?**
 A) Password theft
 B) Reverse engineering
 C) Data theft
 D) Wireless vulnerability
7. **Hackers cannot do which of the following after compromising your phone?**
 A) Steal your information
 B) Rob your e-money
 C) Spying
 D) Shoulder surfing
8. **Which of the following is not an OS for mobile?**
 A) Palm
 B) Mango
 C) Microsoft office
 D) Android
9. _____ works in background and steals sensitive data.
 A) Virus
 B) Shareware
 C) Adware
 D) Trojan
10. **Trojans do not do one of the following. Identify that.**
 A) Deleting Data
 B) Protecting Data
 C) Modifying Data
 D) Copying Data
11. **A computer _____ is a malicious code which self-replicates by copying itself to other programs.**
 A) program
 B) application
 C) worm
 D) virus
12. _____ gets installed & stays hidden in your computer's memory. It stays involved to the specific type of files which it infects.
 A) Boot Sector Virus
 B) Direct Action Virus
 C) Multipartite Virus
 D) Polymorphic Virus

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6. a) Write a note each of the following:

- i) Boot Sector viruses
- ii) Program Viruses
- iii) Stealth Viruses
- iv) Polymorphic viruses.

b) Explain any four types of Identity Theft.

8	L1	3	1
8	L1	4	1

Unit – III

7. a) List forensics life cycle phases. Explain the first two phases.

8	L1	5	1
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b) What is chain of custody? Why is it used? What are the documents that chain of custody must include?

8	L1	5	1
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8. a) List any six security issues that are associated with social networking sites. Give any 3 examples of security measures used by social networking sites.

8	L1	5	1
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b) What does testify in digital forensics involve? When can a witness testify evidence? What are the principles applied to maintain the integrity of digital evidence?

8	L1	5	1
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

13. In _____ some cyber-criminals redirect the legitimate users to different phishing sites and web pages via emails, IMs, ads and spyware.
 A) Phishing B) MiTM attack
 C) URL Redirection D) DoS
14. Phishers often develop _____ websites for tricking users & filling their personal data.
 A) legitimate B) illegitimate
 C) genuine D) official
15. Which of the following type of data, phishers cannot steal from its target victims?
 A) bank details B) phone number
 C) passwords D) apps installed in the mobile
16. _____ type of phishing became very popular as if it has been sent from a legitimate source with a legitimate link to its official website.
 A) Algo-based phishing B) Email-based phishing
 C) Domain Phishing D) Vishing
17. _____ is hiding of data within data, where we can hide images, text, and other messages within images, videos, music or recording files.
 A) Cryptography B) Steganography
 C) Tomography D) Chorography
18. The _____ model is 7-layer architecture where each layer is having some specific functionality to perform.
 A) TCP/IP B) OIS
 C) Cloud D) OSI
19. _____ is a popular command-line packet analyser.
 A) Wireshark B) Metasploit
 C) Snort D) Tcpdump
20. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 A) Dark web B) World Wide Web
 C) Haunted web D) Surface web

PART - B: DESCRIPTIVE ANSWER QUESTIONS.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is cyber stalking? How cyber stalking works?	8	L*1	1	1
	b) Discuss the types and techniques of Credit Card Frauds.	8	L1	1	1
2.	a) What is Reconnaissance? Discuss passive and active Reconnaissance.	8	L1	1	1
	b) Define the terms: Mishing, Vishing and Smishing. Explain how Vishing is carried out.	8	L1	2	1
3.	a) What is social engineering? Briefly explain the two types of social engineering.	8	L1	1	1
	b) List and explain any 5 Bluetooth hacking tools. Explain Bluejacking and Bluesnarfing.	8	L1	2	1
Unit – II					
4.	a) What are Proxy servers and Anonymizers? List the purposes of a Proxy server.	8	L1	3	1
	b) Explain the following Phishing methods: i) Dragnet ii) Lobsterpot.	8	L1	4	1
5.	a) What is DoS attack? What can it do? Discuss the classification of DoS attacks.	8	L1	3	1
	b) List and explain any five ID theft counter measures.	8	L1	4	1